

EXPERIMENT 1

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Semester: 5

Subject Name: Computer Network

Subject Code: 24CSH-337

Experiment 2 – Study the basic network command and Network configuration commands.

Aim

Study the basic network command and Network configuration commands like ping, variations of ip config, tracert, nslookup, netstat, arp, rarp, hostname, pathping and basic networking commands.

Objective

- 1. To understand the purpose and usage of basic network diagnostic and configuration commands in Windows 11.**
- 2. To study the working of networking commands such as Ping, IPConfig, Tracert, NSLookup, Netstat, ARP, Hostname, PathPing, Route, and Net Use.**
- 3. To learn how these commands are used for network configuration, connectivity testing, and troubleshooting.**
- 4. To interpret the output of various networking commands for analyzing and diagnosing network-related issues.**

Software

- Windows PowerShell
- Windows Operating System

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Practical / Experiment Steps Step

-1

Open Windows PowerShell

Step 2

Type Various Commands and analyse their output.

1 – ping command

A) ping www.google.com

```
PS C:\Users\Jagrath> ping -f 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=25ms TTL=118
Reply from 8.8.8.8: bytes=32 time=20ms TTL=118
Reply from 8.8.8.8: bytes=32 time=21ms TTL=118
Reply from 8.8.8.8: bytes=32 time=234ms TTL=118

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 20ms, Maximum = 234ms, Average = 75ms
```

B) ping -n 2 www.google.com

```
PS C:\Users\Jagrath> ping -l 64 8.8.8.8

Pinging 8.8.8.8 with 64 bytes of data:
Reply from 8.8.8.8: bytes=64 time=253ms TTL=118
Reply from 8.8.8.8: bytes=64 time=233ms TTL=118
Reply from 8.8.8.8: bytes=64 time=22ms TTL=118
Reply from 8.8.8.8: bytes=64 time=21ms TTL=118

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 21ms, Maximum = 253ms, Average = 132ms
```

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C) ping -l 64 8.8.8.8

```
PS C:\Users\Jagrath> ping -n 2 www.google.com

Pinging www.google.com [142.251.152.119] with 32 bytes of data:
Reply from 142.251.152.119: bytes=32 time=282ms TTL=119
Reply from 142.251.152.119: bytes=32 time=27ms TTL=119

Ping statistics for 142.251.152.119:
    Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 27ms, Maximum = 282ms, Average = 154ms
```

D) ping -f 8.8.8.8

```
PS C:\Users\Jagrath> ping www.google.com

Pinging www.google.com [142.251.152.119] with 32 bytes of data:
Reply from 142.251.152.119: bytes=32 time=242ms TTL=119
Reply from 142.251.152.119: bytes=32 time=21ms TTL=119
Reply from 142.251.152.119: bytes=32 time=233ms TTL=119
Reply from 142.251.152.119: bytes=32 time=64ms TTL=119

Ping statistics for 142.251.152.119:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 21ms, Maximum = 242ms, Average = 140ms
```

2 – ipconfig

A) ipconfig

```
PS C:\Users\Jagrath> ipconfig /all

Windows IP Configuration

    Host Name . . . . . : Jaguuu
    Primary Dns Suffix . . . . . :
    Node Type . . . . . : Hybrid
    IP Routing Enabled. . . . . : No
    WINS Proxy Enabled. . . . . : No

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :
    Description . . . . . : Intel(R) Ethernet Connection (16) I219-V
```

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B) ipconfig /all

```
PS C:\Users\Jagrath> ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:
```

C) ipconfig /release

D) ipconfig /renew

-Note: commands C) and D) release and get the new IP address respectively and may output a error if not all network media are connected, this error can be ignored.

3- tracert

A) tracert 8.8.8.8

```
PS C:\Users\Jagrath> tracert -h 2 8.8.8.8

Tracing route to dns.google [8.8.8.8]
over a maximum of 30 hops:

  0  18 ms  5 ms  6 ms  10.18.159.16
  1  155 ms  6 ms  5 ms  dsldevice.lan [192.168.1.254]

Trace complete.
```

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B) `tracert -h 8.8.8.8`

```
PS C:\Users\Jagrath> tracert 8.8.8.8

Tracing route to dns.google [8.8.8.8]
over a maximum of 30 hops:

  1    25 ms    14 ms    6 ms    10.18.159.16
  2    29 ms     8 ms    8 ms    dsldevice.lan [192.168.1.254]
  3    18 ms    18 ms    15 ms    100.94.0.1
  4    36 ms    23 ms    11 ms    192.168.201.60
  5     *        *        *        Request timed out.
  6    22 ms    21 ms   141 ms    192.178.83.35
  7    26 ms    32 ms    23 ms    142.251.52.213
  8   254 ms    21 ms    21 ms    dns.google [8.8.8.8]

Trace complete
```

C) `tracert -d 8.8.8.8`

```
PS C:\Users\Jagrath> tracert -d 8.8.8.8

Tracing route to 8.8.8.8 over a maximum of 30 hops

  1    15 ms     3 ms     2 ms    10.18.159.16
  2    23 ms     9 ms     7 ms    192.168.1.254
  3    24 ms    11 ms    10 ms    100.94.0.1
```

4- nslookup

A) nslookup www.google.com

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```
PS C:\Users\Jagrath> nslookup www.google.com
Server: UnKnown
Address: 10.18.159.16

Non-authoritative answer:
Name: www.google.com
Addresses: 2001:4860:482b:7700::
           2001:4860:482d:7700::
           2001:4860:482a:7700::
           2001:4860:4826:7700::
           2001:4860:4829:7700::
           2001:4860:4827:7700::
           2001:4860:4828:7700::
           2001:4860:482c:7700::
           142.251.157.119
           142.251.154.119
```

B) nslookup 8.8.8.8

```
PS C:\Users\Jagrath> nslookup 8.8.8.8
Server: UnKnown
Address: 10.18.159.16

Name: dns.google
Address: 8.8.8.8
```

C) nslookup -type=A 8.8.8.8

```
PS C:\Users\Jagrath> nslookup -type=A 8.8.8.8
Server: UnKnown
Address: 10.18.159.16

Name: dns.google
Address: 8.8.8.8
```

D) nslookup -type=AAAA 8.8.8.8

```
PS C:\Users\Jagrath> nslookup -type=AAAA 8.8.8.8
Server: UnKnown
Address: 10.18.159.16

Name: dns.google
Address: 8.8.8.8
```

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5- netstat A)

netstat

```
PS C:\Users\Jagrath> netstat
```

Active Connections

Proto	Local Address	Foreign Address	State
TCP	10.18.159.98:49410	4.213.25.241:https	ESTABLISHED
TCP	10.18.159.98:52641	lb-140-82-114-26-iad:https	ESTABLISHED
TCP	10.18.159.98:55566	104.18.39.21:https	ESTABLISHED
TCP	10.18.159.98:56398	104.20.28.164:https	ESTABLISHED
TCP	10.18.159.98:58893	52.123.176.97:https	ESTABLISHED
TCP	10.18.159.98:58898	4.213.25.241:https	ESTABLISHED
TCP	10.18.159.98:59679	52.123.173.252:https	ESTABLISHED
TCP	10.18.159.98:60547	ec2-18-97-36-56:https	ESTABLISHED
TCP	10.18.159.98:62200	172.64.155.209:https	ESTABLISHED
TCP	10.18.159.98:64391	104.21.15.154:https	ESTABLISHED
TCP	127.0.0.1:49679	Jaguuu:49680	ESTABLISHED
TCP	127.0.0.1:49680	Jaguuu:49679	ESTABLISHED
TCP	127.0.0.1:49681	Jaguuu:49682	ESTABLISHED
TCP	127.0.0.1:49682	Jaguuu:49681	ESTABLISHED
TCP	127.0.0.1:49697	Jaguuu:49698	ESTABLISHED
TCP	127.0.0.1:49698	Jaguuu:49697	ESTABLISHED
TCP	127.0.0.1:49699	Jaguuu:49700	ESTABLISHED

B) netstat -n

```
PS C:\Users\Jagrath> netstat -n
```

Active Connections

Proto	Local Address	Foreign Address	State
TCP	10.18.159.98:49410	4.213.25.241:443	ESTABLISHED
TCP	10.18.159.98:55566	104.18.39.21:443	ESTABLISHED
TCP	10.18.159.98:55725	140.82.114.25:443	ESTABLISHED
TCP	10.18.159.98:56398	104.20.28.164:443	ESTABLISHED
TCP	10.18.159.98:58893	52.123.176.97:443	ESTABLISHED
TCP	10.18.159.98:58898	4.213.25.241:443	ESTABLISHED
TCP	10.18.159.98:59679	52.123.173.252:443	ESTABLISHED
TCP	10.18.159.98:63328	18.97.36.64:443	ESTABLISHED
TCP	10.18.159.98:64391	104.21.15.154:443	ESTABLISHED
TCP	127.0.0.1:49679	127.0.0.1:49680	ESTABLISHED
TCP	127.0.0.1:49680	127.0.0.1:49679	ESTABLISHED
TCP	127.0.0.1:49681	127.0.0.1:49682	ESTABLISHED
TCP	127.0.0.1:49682	127.0.0.1:49681	ESTABLISHED
TCP	127.0.0.1:49697	127.0.0.1:49698	ESTABLISHED
TCP	127.0.0.1:49698	127.0.0.1:49697	ESTABLISHED
TCP	127.0.0.1:49699	127.0.0.1:49700	ESTABLISHED
TCP	127.0.0.1:49700	127.0.0.1:49699	ESTABLISHED
TCP	127.0.0.1:49701	127.0.0.1:49702	ESTABLISHED

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C) netstat -p TCP

```
PS C:\Users\Jagrath> netstat -p TCP

Active Connections

Proto Local Address           Foreign Address         State
TCP   10.18.159.98:49410       4.213.25.241:https      ESTABLISHED
TCP   10.18.159.98:50549       a23-55-106-105:https    ESTABLISHED
TCP   10.18.159.98:50550       57.155.101.212:https    TIME_WAIT
TCP   10.18.159.98:50552       a2-20-157-120:http      TIME_WAIT
TCP   10.18.159.98:50553       tzdelb-bk-in-f3:http    TIME_WAIT
TCP   10.18.159.98:50554       a23-55-106-98:http      TIME_WAIT
TCP   10.18.159.98:50555       199.232.214.172:http    TIME_WAIT
TCP   10.18.159.98:50556       20.190.146.36:https     ESTABLISHED
TCP   10.18.159.98:50557       20.190.146.36:https     ESTABLISHED
TCP   10.18.159.98:50558       57.155.101.212:https    TIME_WAIT
TCP   10.18.159.98:50559       57.155.101.212:https    TIME_WAIT
TCP   10.18.159.98:50561       57.155.101.212:https    TIME_WAIT
TCP   10.18.159.98:50562       57.155.101.212:https    TIME_WAIT
TCP   10.18.159.98:50566       server-54-192-142-12:https TIME_WAIT
```

D) netstat -p UDP

6-arp

A) arp -a

```
PS C:\Users\Jagrath> arp -a

Interface: 10.18.159.98 --- 0xa
Internet Address      Physical Address        Type
10.18.159.16          56-f8-06-f2-24-db      dynamic
10.18.159.255         ff-ff-ff-ff-ff-ff      static
224.0.0.22            01-00-5e-00-00-16      static
224.0.0.251           01-00-5e-00-00-fb      static
224.0.0.252           01-00-5e-00-00-fc      static
255.255.255.255       ff-ff-ff-ff-ff-ff      static
```

B) arp -d *

_Note: arp -d will delete the entries so don't run this command unless you REALLY know what you are doing.

7 – hostname

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```
PS C:\Users\Jagrath> hostname  
Jaguuu
```

8- pathping

```
Computing statistics for 275 seconds...
Hop  RTT      Source to Here   This Node/Link   Address
  0                               Lost/Sent = Pct  Lost/Sent = Pct  Jaguuu [10.18.159.98]
  1  25ms      0/ 100 = 0%      0/ 100 = 0%      |
  2  68ms      1/ 100 = 1%      1/ 100 = 1%      dsldevice.lan [192.168.1.254]
  3  44ms      2/ 100 = 2%      2/ 100 = 2%      |
  4  45ms      0/ 100 = 0%      0/ 100 = 0%      100.94.0.1
  5  41ms      0/ 100 = 0%      0/ 100 = 0%      |
  6  39ms      2/ 100 = 2%      2/ 100 = 2%      192.168.201.60
  7  43ms      0/ 100 = 0%      0/ 100 = 0%      |
  8  58ms      0/ 100 = 0%      0/ 100 = 0%      192.168.200.1
  9  66ms      1/ 100 = 1%      1/ 100 = 1%      |
 10  66ms      0/ 100 = 0%      0/ 100 = 0%      121.23.41.103.netplus.co.in [103.41.23.121]
 11  60ms      0/ 100 = 0%      0/ 100 = 0%      |
 12  66ms      0/ 100 = 0%      0/ 100 = 0%      aes-static-125.34.144.59.airtel.in [59.144.34.125]
 13  58ms      0/ 100 = 0%      0/ 100 = 0%      |
 14  66ms      1/ 100 = 1%      1/ 100 = 1%      aes-static-125.34.144.59.airtel.in [59.144.34.125]
 15  66ms      0/ 100 = 0%      0/ 100 = 0%      |
 16  66ms      0/ 100 = 0%      0/ 100 = 0%      116.119.161.26
 17  66ms      0/ 100 = 0%      0/ 100 = 0%      |
 18  66ms      0/ 100 = 0%      0/ 100 = 0%      72.14.243.0
 19  60ms      0/ 100 = 0%      0/ 100 = 0%      |
 20  60ms      0/ 100 = 0%      0/ 100 = 0%      142.251.154.119
```

Learning Outcome

After completing this experiment, the learner will be able to:

1. Execute basic network configuration and diagnostic commands in a Windows 11 environment.
2. Interpret the output of commands such as **Ping, IPConfig, Tracert, NSLookup, Netstat, ARP, Hostname, PathPing, Route, and Net Use.**
3. Identify and troubleshoot basic network connectivity and configuration issues using command-line tools.
4. Explain the purpose and practical applications of commonly used networking commands.